

3] Explain the role of clauses in SQL statements.

Ans:

Common Clauses and Their Roles

1. SELECT Clause:

Specifies the columns to retrieve from a table.

Example: `SELECT student name, age FROM students;`

2. FROM Clause:

Identifies the table or tables from which to retrieve data.

Example: `SELECT * FROM students;`

3. WHERE Clause:

Filters rows based on specified conditions.

Example: `SELECT * FROM students WHERE age > 15;`

4. GROUP BY Clause:

Groups rows with the same values in specified columns for aggregation.

Example: `SELECT class, COUNT (*) FROM students GROUP BY class;`

5. HAVING Clause:

Filters groups created by the GROUP BY clause based on a condition.

Example: `SELECT class, COUNT (*) FROM students GROUP BY class
HAVING COUNT(*) > 2;`

6. ORDER BY Clause:

Sorts the query result by specified columns in ascending or descending order.

Example: `SELECT * FROM students ORDER BY age DESC;`

7. LIMIT Clause (or TOP in some databases):

Restricts the number of rows returned.

Example: `SELECT * FROM students LIMIT 5;`